Design and Analysis of Active High Pass and Low Pass Filters for an Audio Analyzer

Boyi Dai

I. High and Low Pass Filters

Filtering is an important component in signal processing systems. In this project, both a high pass filter and a low pass filter were implemented to control the frequency content of an input signal within the audio analyzer. The high pass filter attenuates low frequency components while allowing higher frequency signals to pass, whereas the low pass filter attenuates high frequency components. Together these filters ensure that only the relevant portion of the signal spectrum is analyzed while unwanted frequency components are suppressed.

Both filters were implemented using the Sallen–Key active filter topology described in [1]. This topology uses an operational amplifier along with resistors and capacitors to produce a second order filter response. The active design provides buffering and allows the cutoff frequency and filter behavior to be adjusted through component selection.

Fig. 3 shows the full schematic of the audio analyzer system, which includes the high pass and low pass filter stages used to shape the frequency response of the signal.

A. Design and Analysis

The low pass (LP) and high pass (HP) filters were implemented using the Sallen-Key topology described in [1]. In the general Sallen-Key configuration, the op amp stage can provide gain $K = 1 + \frac{R_4}{R_3}$. However, in this design the op amp is configured as a unity gain buffer, so $K = 1$. Substituting $K = 1$ simplifies the general Sallen-Key design equations.

For both filters, the corner frequency is determined by the resistor and capacitor values according to

$$f_c = \frac{1}{2\pi\sqrt{R_1R_2C_1C_2}} \tag{1}$$

where R_1 and R_2 are the resistors in the filter network, C_1 and C_2 are the capacitors, and f_c is the filter corner frequency.

The sharpness of the filter response is determined by the quality factor Q, which depends on the component values of each stage.

For the low pass filter, the quality factor is

$$Q_{LP} = \frac{\sqrt{R_6R_7C_7C_8}}{R_6C_7 + R_7C_7} \tag{2}$$

This is the Sallen-Key report for ECE 2600 Electronics at the University of Virginia

For the high pass filter, the quality factor is

$$Q_{HP} = \frac{\sqrt{R_8R_9C_9C_{10}}}{R_9C_9 + R_9C_{10}} \tag{3}$$

where R_6, R_7, R_8, and R_9 are the resistors and C_7, C_8, C_9, and C_{10} are the capacitors from Fig. 3.

Component values were selected using a Python script that searched available resistor and capacitor values from the studio kit. Because the available component values were limited to those provided in the studio kit, the exact target cutoff frequencies and quality factor could not always be achieved simultaneously. The script selected component combinations that minimized the error between the desired and achievable cutoff frequencies while keeping the quality factor close to $Q = 0.707$, which provides a maximally flat passband without gain peaking. The final component values used are the best trade off between these constraints and the desired filter performance.

For the high pass filter, the selected component values were

$$C_9 = C_{10} = 470\,\text{pF}, \quad R_8 = 1\,\text{M}\Omega, \quad R_9 = 470\,\text{k}\Omega$$

These values produce a corner frequency of 493.9 Hz and a quality factor of $Q_{HP} = 0.729$.

For the low pass filter, the selected component values were

$$C_7 = 680\,\text{pF}, \quad C_8 = 1.5\,\text{nF}, \quad R_6 = 1\,\text{M}\Omega, \quad R_7 = 560\,\text{k}\Omega$$

These values produce a corner frequency of 210.6 Hz and a quality factor of $Q_{LP} = 0.712$.

The resulting corner frequencies are separated by more than 200 Hz, satisfying the design requirements.

B. Simulation and Experimental Results

The filter designs were verified using both circuit simulation and experimental measurements to confirm that the implemented circuits produced the expected cutoff frequencies.

1) Simulation Setup: AC signal simulations were performed in KiCad using the SPICE simulator [2]. The input source was configured with an AC magnitude of 1 V, and the output magnitude V_{out} was measured relative to the input to generate Bode magnitude plots. The op amp was powered with ± 5 V supply rails and configured as a unity gain buffer ($K = 1$).

2) Experimental Measurement Setup: Experimental measurements were performed using the Analog Discovery waveform generator and network analyzer [3]. A 1 V sinusoidal input was applied and the frequency response was measured using a logarithmic frequency sweep.

3) Results: Figure 1 shows the high pass filter frequency response with KiCad simulation and WaveForms measurement overlaid and Figure 2 shows the low pass filter frequency response with KiCad simulation and WaveForms measurement overlaid.

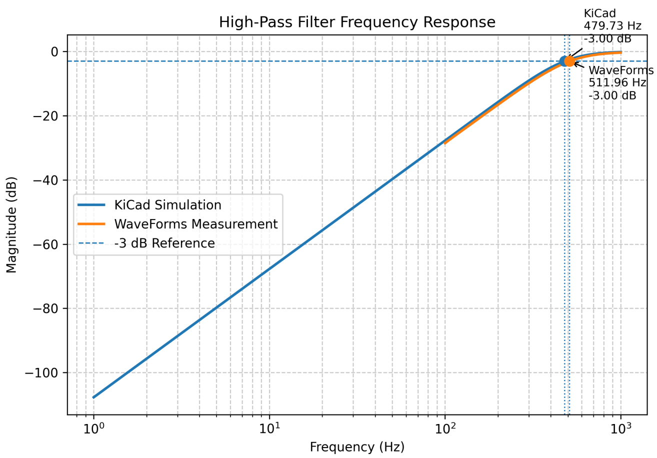

Fig. 1. High pass filter frequency response showing KiCad simulation and WaveForms measurement.

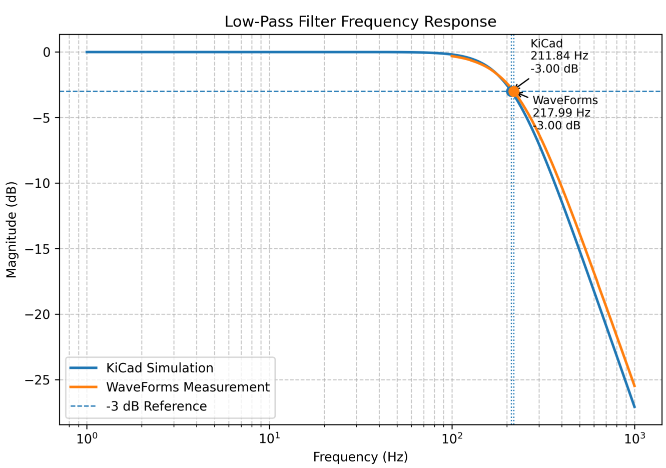

Fig. 2. Low pass filter frequency response showing KiCad simulation and WaveForms measurement.

4) Discussion: The results confirm that the selected resistor and capacitor values successfully produce the desired filtering behavior. For the high pass filter, the theoretical cutoff frequency was 493.9 Hz while the experimentally measured cutoff frequency was 511.96 Hz, resulting in a percentage error of approximately 3.66%. For the low pass filter, the theoretical cutoff frequency was 210.6 Hz while the measured cutoff frequency was 217.99 Hz, corresponding to a percentage error of approximately 3.51%.

These relatively small errors indicate that the theoretical design equations and the measured circuit behavior are accurate. Deviations of a few percent can be caused by component tolerances, resistances in the physical implementation, and non ideal characteristics of the operational amplifier. Despite these small discrepancies, the measured responses closely match the predicted behavior, demonstrating that the implemented filters operate as designed and effectively separate the desired frequency bands for the audio analyzer system.

REFERENCES

[1] *Analysis of the sallen-key architecture (sloa024b)*, Texas Instruments Incorporated, Sep. 2002. [Online]. Available: https://www.ti.com/lit/an/sloa024b/sloa024b.pdf.

[2] *Kicad eda software suite*, KiCad. [Online]. Available: https://www.kicad.org/.

[3] *Analog discovery 2*, Digilant. [Online]. Available: https://digilent.com/reference/test-and-measurement/analog-discovery-2/start.

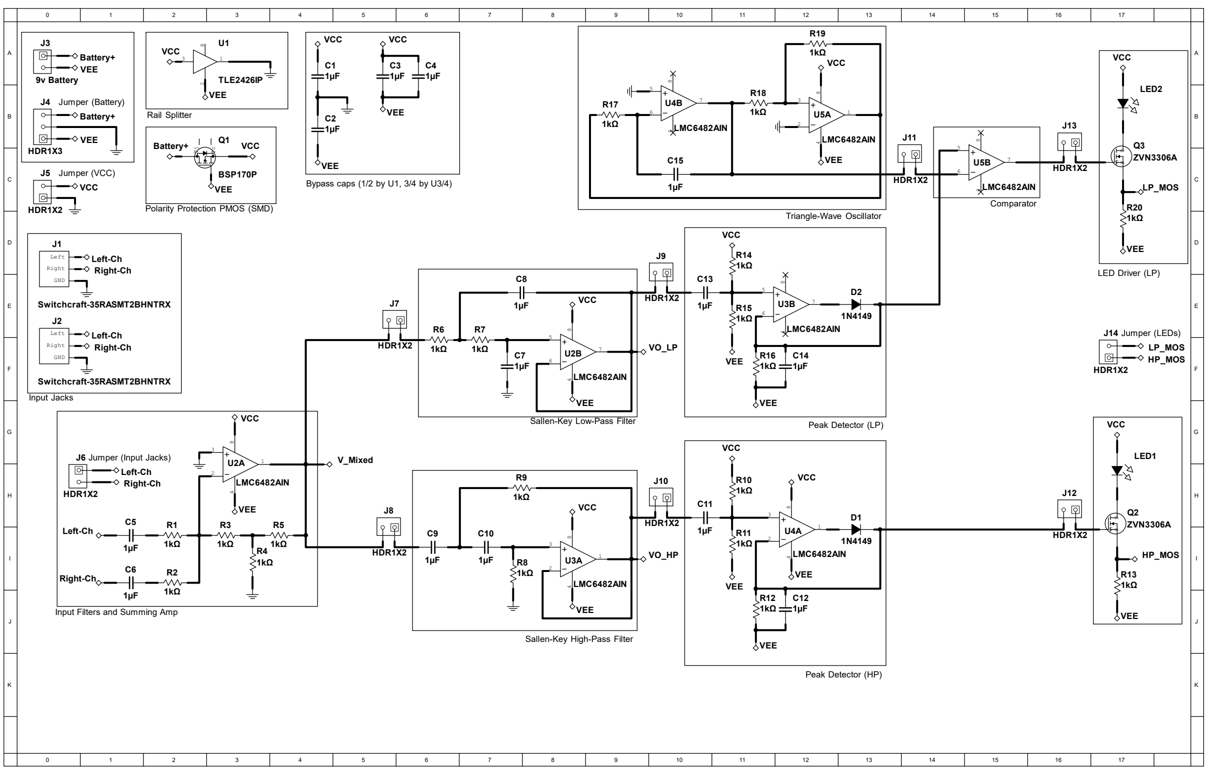

Fig. 3. Full schematic of the audio analyzer showing the high pass and low pass filter stages used in the design.